

Optical Intelligent Reflecting Surface-Assisted Visible Light Covert Communication With NOMA: An Effective Covert Spectral Efficiency Perspective

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Abstract—Covert communications play a crucial role in modern communication systems, ensuring privacy and security in urban environments. Visible light communication (VLC) emerges as a promising technology due to its potential for covert operations within established infrastructures. In this paper, the covert communication for non-orthogonal multiple access (NOMA)-based VLC systems is investigated, by leveraging the assistance of optical intelligent reflecting surface (OIRS). We develop a system model that allows covert user information to be hidden in the public user information, and to disrupt the warden by altering the channel gain through OIRS. Moreover, the problem is formulated to maximize the effective covert spectral efficiency (SE) of the visible light covert communication (VLCC) system with the detection error probability (DEP) and connection outage probability (COP) constraints, and then an optimization framework is proposed to solve the non-convex problem for both multiple-input single-output (MISO) and single-input single-output (SISO) scenarios. For MISO, a block coordinate descent-based optimization algorithm is developed, decomposing the problem into two sub-problems, named power allocation and OIRS configuration sub-problems. For SISO, closed-form solutions for the power ratio relative to OIRS configuration are derived, simplifying the optimization process. Moreover, we analyzed two distinct cases where Willie either possesses or lacks the knowledge of

OIRS state, thereby revealing the impact of warden capabilities on covert communications. Furthermore, the effect of imperfect acquisition of Willie's channel state information is also examined on the effective covert SE. Simulation results demonstrate that the proposed algorithm enhances the effective covert SE of VLCC systems, while exploring the impact of OIRS units number and other key factors. These findings highlight the potential of OIRS to improve the covert performance in future optical communications.

Index Terms—Optical intelligent reflecting surface (OIRS), covert communication, non-orthogonal multiple access (NOMA), visible light communication (VLC), detection error probability (DEP), connection outage probability (COP).

I. INTRODUCTION

COVERT communications, also known as low-probability-of-detection (LPD) communications, have gained significant attention in recent years due to its critical role in ensuring privacy and security in modern communication systems [2]. As wireless communication networks continue to evolve, the increasing threats of eavesdropping and unauthorized access have underscored the importance of developing methods to conceal transmitted information [3], [4]. Covert communications aim to minimize the probability of detection by unauthorized entities, making it a fundamental aspect of secure communication in sensitive environments [5], [6].

Building upon this requirement, non-orthogonal multiple access (NOMA) has emerged as an effective technique that provides opportunities for covert transmissions and simultaneously improves the spectral efficiency (SE), particularly at high signal-to-noise ratio (SNR) [7]. By exploiting power-domain multiplexing, NOMA encodes covert information within the power levels of public signals [8]. For instance, the authors in [9] enables the simultaneous transmission of public and private information, thereby enhancing the covert transmission rate (CTR). Existing studies show that adaptive power allocation can effectively obscure the signal of covert user within the public transmission [10], making the warden's detection increasingly difficult [11]. This line of research demonstrates that the principle of NOMA, which leverages superposition coding and successive interference cancellation (SIC) [12], inherently supports covert communications [13].

Received 6 August 2025; revised 30 November 2025; accepted 22 January 2026. Date of current version 5 February 2026. This work was supported in part by the Collaborative Innovation Special Project of Shenzhen Municipal Government under Grant CJGJZD20230724092959002, in part by NSF under Grant ECCS-2302469, and in part by Japan Science and Technology Agency (JST) Adopting Sustainable Partnerships for Innovative Research Ecosystem (ASPIRE) under Grant JPMJAP2326. An earlier version of this paper was presented at IEEE GLOBECOM in 2025 [1]. The associate editor coordinating the review of this article and approving it for publication was S. De. (Corresponding author: Fang Yang.)

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Digital Object Identifier 10.1109/TWC.2026.3658113

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TABLE I
WORKS ON COVERT COMMUNICATIONS

Reference	Band	Scenario	Detector	Metric	MA Scheme	(O)IRS	Randomness Source
[3]	RF	SISO	Energy Detector	Detection Error Probability	✗	✗	Channel Fading
[4]	RF	SISO	Energy Detector	Effective CTR	✗	✗	Channel Fading
[5]	RF	MISO	Likelihood Ratio Test	Effective CTR	✗	✗	Channel Fading
[6]	RF	MISO	Likelihood Ratio Test	SNR	✗	✗	Environmental Noise
[8]	RF	SISO	Energy Detector	Average CTR	NOMA	✗	Channel Fading
[9]	RF	SISO	Energy Detector	Effective CTR	NOMA	✓	Channel Fading
[10]	RF	SISO	Likelihood Ratio Test	Allowable Transmit Power	✗	✗	Random Transmit Power
[11]	RF	SISO for UAV	Energy Detector	Average CTR	✗	✗	Position Uncertainty
[12]	RF	SISO	Energy Detector	Effective CTR	NOMA	✗	Random Transmit Power
[13]	RF	SISO	Energy Detector	Effective CTR	NOMA	✗	Channel Fading
[14]	RF	SISO	Energy Detector	Effective CTR	NOMA	✓	Channel Fading
[15]	RF	SISO	Energy Detector	Effective CTR	NOMA	✓	IRS Uncertainty
[23]	RF	SISO	Likelihood Ratio Test	SNR	✗	✓	Environmental Noise
[18]	VLC	SISO	Likelihood Ratio Test	Mutual Information	✗	✗	Signal Distribution
[20]	VLC	SISO for UAV	Likelihood Ratio Test	Mutual Information	✗	✗	Signal Distribution
[22]	VLC	SISO for UAV	Energy Detector	Effective CTR	NOMA	✗	Environmental Noise
[24]	VLC	SISO	Energy Detector	Effective CTR	✗	✓	Environmental Noise
Our Work	VLC	MISO / SISO	Energy Detector	Effective CTR	NOMA	✓	Random Transmit Power

To further enhance covertness, recent studies have incorporated intelligent reflecting surfaces (IRS), which introduce an additional degree of freedom by reconfiguring the wireless channel [14]. Moreover, the adaptive channel manipulation capability of IRS complements the power allocation mechanism of NOMA [15], enabling precise control over the power levels and spatial distribution of the transmitted signals. Furthermore, tile-based IRS systems can partition the surface into multiple tiles, each comprising numerous IRS units [16], thereby simplifying configuration while retaining operational flexibility. To sum up, the adaptability of IRS not only improves the overall system performance but also provides an additional degree of freedom for enhancing covert communications [17]. For instance, by optimizing the phase shifts of the IRS to maximize the effective CTR, an additional stochastic characteristic is introduced into the propagation from the warden's perspective, as demonstrated in [15]. Besides, the authors in [14] derive the performance impact of combined IRS-NOMA on covert communication systems.

In parallel, visible light communication (VLC) has gained traction as a natural candidate for covert communications due to its distinctive features [18]. Unlike traditional radio frequency (RF) communications, VLC operates in the visible light spectrum, leveraging light-emitting diodes (LEDs) for data transmission [19]. This distinction makes VLC a suitable candidate for secure indoor and urban communications, as the optical signal can not penetrate walls, significantly reducing the risk of interception from external eavesdroppers [19]. Furthermore, VLC provides a wide unlicensed spectrum, making it an efficient and secure alternative to RF-based communication systems [19]. These advantages position VLC as a critical technology for achieving covert and private communication in densely populated environments. Recently, significant research has been conducted for the visible light covert communication (VLCC) system. The derivation of the

optimal signal distribution based on the real and non-negative characteristics of VLC signal is formulated in [18], and the fundamental performance limit of VLCC systems is also studied. Moreover, uncrewed aerial vehicles (UAVs) for VLCC systems have also been studied, including the optimization of the UAV altitude [20] as well as the peak optical intensity [21], and the power allocation under covert communication constraints [22].

To further improve the performance of the VLC system, the optical intelligent reflecting surface (OIRS) is introduced [23], [25]. Specifically, OIRS has many advanced optical units capable of dynamically modifying the properties of the optical propagation channel [23]. By intelligently reconfiguring the units, OIRS can change channel conditions and enhance the signal quality for intended users [9], [26]. For instance, the authors in [24] implemented the VLCC utilizing OIRS and maximized the effective CTR. However, the optimization and research methods for IRS in RF domain cannot be directly adopted for OIRS in VLC, because the characteristics of optical signals and the implementation of OIRS are different from traditional RF communications [27], [28]. Currently, OIRS hardware architectures can be categorized into two main approaches, namely the mirror array-based OIRS and the metasurface-based OIRS [29]. Meta-surfaces can be manufactured to have specific electromagnetic properties [30], enabling precise control over the amplitude and direction of reflected light [31]. For the mirror-based OIRS, it is easier to fabricate and do not require advanced nanofabrication techniques, making it more cost-effective and accessible. Therefore, this paper focuses on the investigation of OIRS implemented through mirror arrays, whose channel characteristics are more conducive to analysis [27].

To sum up, the combination of OIRS and NOMA has strong potential to simultaneously configure optical channels and embed covert signals in the power domain, yet its benefits

remain largely unexplored. Therefore, the OIRS-aided VLCC with NOMA is studied in both multiple-input single-output (MISO) and single-input single-output (SISO) scenarios, with the goal of maximizing the effective covert spectral efficiency (SE). Specifically, we analyze the channel models of both the line-of-sight (LoS) and the OIRS-reflected paths, and subsequently establish a corresponding covert transmission scheme. A comprehensive comparison of related covert communication works is provided in Table I, and the main contributions of this paper are summarized as follows:

- A covert communication system for the NOMA-based VLC with OIRS is established, and then the OIRS configuration is represented by a binary matrix based on the analysis of channel gains. Moreover, the detection performance of the warden is discussed, including his decision threshold and the actual detection error probability (DEP), followed by the study of the connection outage probability (COP) for legitimate users. Additionally, we discussed the different capabilities of warden and the impact of its channel state information (CSI) on the system performance. According to the performance evaluation, the objective function for the NOMA-based VLCC system with OIRS is formulated into an effective covert SE maximization problem, which is subject to various constraints, including both the DEP of warden and the COP of public user.
- The effective covert SE maximization problem in MISO scenario is decomposed into two sub-problems according to block coordinate descent (BCD) algorithm, which is subject to the power allocation and OIRS configuration sub-problems. Then, these two sub-problems are iteratively addressed by convex optimization and relaxed minorization-maximization (MM) method, respectively. For the SISO scenario, we derive closed-form solutions for power allocation ratio concerning OIRS configuration, significantly simplifying the optimization process.
- Theoretical analyses on the convergence and computational complexity of the proposed algorithm are discussed in this paper. Moreover, through extensive simulations, it is demonstrated that the proposed algorithm significantly enhances the effective covert SE of VLCC systems. The impact of key factors is also explored, such as the number of OIRS units, power allocation strategies, and values of the constraints on the covertness performance of VLCC systems. The simulation results highlight the potential of OIRS and NOMA as viable solutions for achieving private and efficient optical covert communications.

The remainder of the paper is organized as follows. The system architecture and channel models of both the LoS and OIRS-reflected paths are discussed in Section II, followed by the covert transmission scheme. Section III provides an analysis of the proposed VLCC system when Willie is unaware of the OIRS status, including the performance of both DEP and COP. In Section IV, the problem is formulated to maximize the effective covert SE, and then the BCD algorithm and the approaches for two sub-problems are proposed, followed by the analysis of convergence and computational complexity.

Then, Section V addresses the implications of Willie's knowledge regarding the OIRS state and the impact of its imperfect CSI on covert communication systems. Subsequently, the special case for SISO scenario is discussed in Section VI. Then, the simulation results are provided in Section VII. Finally, the paper is summarized in Section VIII.

Notations: a , $\mathbf{a}$, and $\mathbf{A}$ represent the scalar value, vector, and matrix, respectively. $\mathbb{R}_+$ and $\mathbb{B}$ denote the positive set of real numbers and the set of binary numbers, respectively, and $\mathcal{A}$ represents the defined set. Moreover, $\|\cdot\|_F$, $\Pr(\cdot)$, $\sqrt{\cdot}$, and $(\cdot)^T$ denote the symbols of Frobenius norm, probability value, square root operator, and transpose, respectively. $\arg \max(\cdot)$ is the argument that make the following expression reach its maximum value, while $\min\{\cdot\}$ represents the minimum value in the brackets.

II. SYSTEM MODEL

A. System Architecture

As shown in Fig. 1, a NOMA-based downlink VLCC system with OIRS is considered, where L LEDs are transmitting information to legitimate users Roy and Bob through the power-domain NOMA, and the LED indices are represented by the set $\mathcal{L}$. In particular, the LEDs openly transmit information to the NOMA-weak user Roy. At the same time, the LEDs attend to conduct covert communications with the NOMA-strong user Bob in the presence of a warden Willie, whose objective is to detect any covert communication occurring between LEDs and Bob through the monitored link. Moreover, an OIRS consisting of N units is mounted on the wall to assist covert communications and improve the communication performance, and the indices of N units are denoted by the set $\mathcal{N}$. Without loss of generality, the locations of LEDs, Bob, Roy, Willie, and OIRS units are assumed to be known by the system controller [32], and it can also obtain the full CSI of the whole VLCC system through various channel estimation methods [33], [34].

B. Channel Model

In the OIRS-aided VLCC system, the transmitted signal arrives at the photodetector (PD) via both LoS and OIRS-reflected paths. In particular, based on the position information of the LEDs and receivers, the LoS channel gain between the l -th LED and receiver i can be expressed by the Lambertian model as [35]

$$h_{l,i} = \frac{A(m+1)}{2\pi d_{l,i}^2} \cos^m(\Theta_{l,i}) T(\Phi_{l,i}) \cos(\Phi_{l,i}) g(\Phi_{l,i}), \quad (1)$$

where $i \in \{R, B, W\}$ denotes one of Roy, Bob, and Willie. m and A are the Lambertian index and the physical area of the PD, respectively. $d_{l,i}$ denotes the distance between the l -th LED and the receiver i . $\Theta_{l,i}$ and $\Phi_{l,i}$ represent the corresponding irradiance and incidence angles between the l -th LED and receiver i , respectively. $T(\Phi_{l,i})$ is the optical filter gain, and $g(\Phi_{l,i})$ denotes the optical concentrator gain. When $0 \leq \Phi_{l,i} \leq \Phi_{\text{FoV}}$, the optical concentrator gain can be given by

$$g(\Phi_{l,i}) = \frac{b^2}{\sin^2(\Phi_{\text{FoV}})}, \quad (2)$$

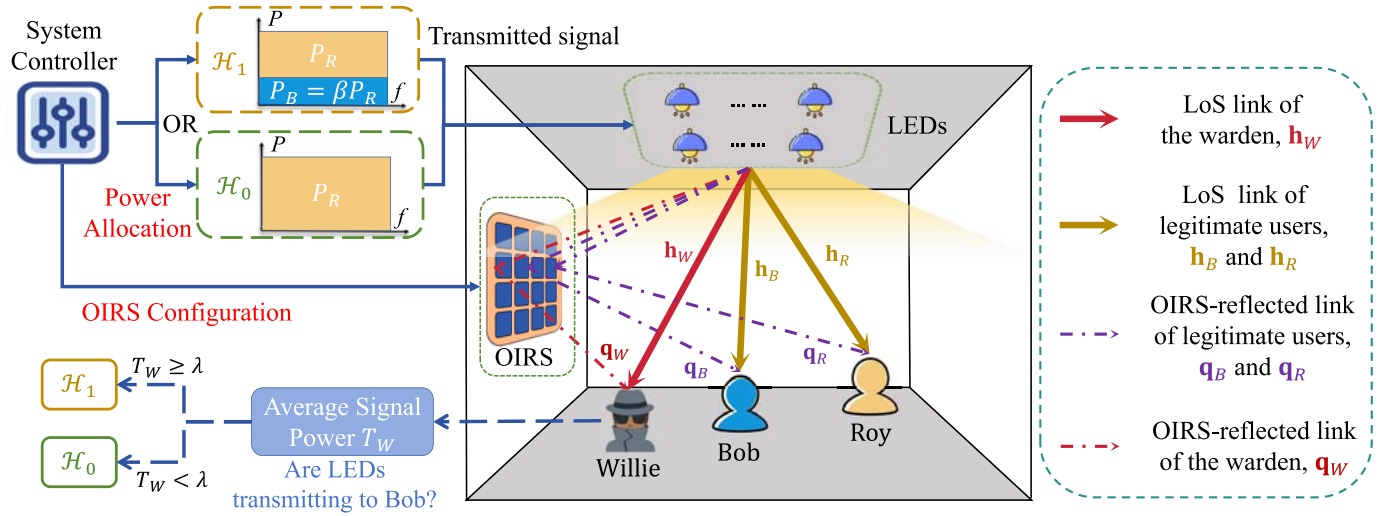


Fig. 1. The system model of the NOMA-based downlink VLCC system with OIRS.

where b and Φ_{FoV} are refractive index and semi-angle of the field-of-view (FoV) of PD, respectively. For ease of discussion, the vector $\mathbf{h}_i = [h_{1,i}, \dots, h_{L,i}]^T \in \mathbb{R}_+^{L \times 1}$ represents the LoS channel gain vector for receiver $i \in \{R, B, W\}$.

For the OIRS-reflected channel in the VLCC system, it is mainly made up of the specular and diffuse reflections. Nevertheless, it has been demonstrated that the gain from diffuse reflection in VLC systems can be over 15 dB lower than that of the specular reflection [25], [35], which indicates that neglecting diffuse reflections will have little effect on the OIRS-reflected channels. Moreover, when studying the specular reflection of the mirror array-based OIRS, the reflected signal can be approximated as the light emitted from a virtual LED in a mirrored position and also follows the Lambertian model by simply adding a reflection loss factor δ [29], [36]. Consequently, the OIRS-reflected channel gain between the l -th LED and receiver $i \in \{R, B, W\}$ through the n -th OIRS unit can be expressed as [29] and [36]

$$f_{l,n,i} = \frac{\delta A(m+1)}{2\pi(d_{l,n} + d_{n,i})^2} \cos^m(\theta_{l,n,i}) T(\phi_{l,n,i}) \cos(\phi_{l,n,i}) g(\phi_{l,n,i}), \quad (3)$$

where $d_{l,n}$ and $d_{n,i}$ denote the distances between the l -th LED and the n -th OIRS unit, the n -th OIRS unit and receiver i , respectively. $\theta_{l,n,i}$ and $\phi_{l,n,i}$ represent the angles of irradiance and incidence for the OIRS-reflected path, respectively. Then, the OIRS-reflected channel gains between the l -th LED and receiver i can be expressed as a vector form $\mathbf{f}_{l,i} = [f_{l,1,i}, \dots, f_{l,N,i}]^T \in \mathbb{R}_+^{N \times 1}$.

For the VLCC system with OIRS, the experimental verification has indicated that the non-line-of-sight (NLoS) channel gain contributes less than 5% of the overall received optical power, which can be ignored in VLC systems [35] or modeled by the CSI uncertainty [27]. Moreover, since the size of an OIRS unit is small compared to the propagation distance, its coverage area is approximately equal to its own physical dimensions [36]. Then, each unit naturally forms a highly directional optical path, which physically limits it to effectively serving a single LED-receiver pair [37]. As a result, each

OIRS unit can establish a reflected channel between only one LED and one receiver [36], thereby a binary vector $s_{l,i} = [s_{l,1,i}, \dots, s_{l,N,i}]^T \in \mathbb{B}^{N \times 1}$, $s_{l,n,i} \in \{0, 1\}$ can be utilized to denote the assignment of OIRS units for the l -th LED and the receiver $i \in \{R, B, W\}$, where $s_{l,n,i} = 1$ represents that the n -th OIRS unit is assigned to receiver i and the l -th LED, and $s_{l,n,i} = 0$ otherwise. Then, the assignment of all OIRS units forms the OIRS configuration matrix $\mathbf{S} \in \mathbb{R}_+^{L \times N \times 3}$. Consequently, the OIRS-reflected channel gains of the receiver $i \in \{R, B, W\}$ can be given by $\mathbf{q}_i = [q_{1,i}, \dots, q_{L,i}]^T \in \mathbb{R}_+^{L \times 1}$, where $q_{l,i} = s_{l,i}^T \mathbf{f}_{l,i}$.

To sum up, the total channel gain vector for the receiver $i \in \{R, B, W\}$ can be obtained by adding the LoS and OIRS-reflected channel gains together, which can be given by

$$\mathbf{g}_i = \mathbf{h}_i + \mathbf{q}_i, \quad i \in \{R, B, W\}. \quad (4)$$

C. Covert Transmission Scheme

In covert communications, it is often necessary to introduce randomness to confuse the warden Willie and reduce the probability of his detection. For public communications between the LEDs and Roy, the random power transmission scheme is exploited, where the transmit power P_R of the LEDs to Roy follows a continuous uniform distribution over the interval $[0, P_{\max}]$, where $P_{\max}$ denotes the maximum transmit power of the LEDs to Roy. Therefore, the probability density function can be given by

$$f_{P_R}(x) = \begin{cases} \frac{1}{P_{\max}}, & 0 \leq x \leq P_{\max}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Then, the transmit powers of all LEDs to Roy can be written into a vector $\mathbf{w}_R \in \mathbb{R}_+^{L \times 1}$. It is worthy noting that if the transmit power to Roy is a constant value, it is straightforward for Willie to flag a covert transmission when an additional power from LEDs is received. Specifically, the introduction of uncertainty in the transmit power leaves Willie unsure about

whether an increase in received power stems from the LEDs-to-Bob transmission or merely reflects variations in transmit power of LEDs to Roy.

Moreover, when the LEDs send a covert message to Bob, each LED needs to send the superimposed signal $\sqrt{P_R}u_R[k] + \sqrt{P_{B,l}}u_B[k]$, $\forall l \in \mathcal{L}$ through the power-domain NOMA, where $\sqrt{P_R}$ and $\sqrt{P_{B,l}}$ are the power assigned to Roy and Bob by the l -th LED, respectively. $u_R[k]$ and $u_B[k]$ denote the k -th normalized symbols of Roy and Bob, respectively. Moreover, the transmit power vector of the LEDs to Bob can be expressed as $\mathbf{w}_B = [\sqrt{P_{B,1}}, \dots, \sqrt{P_{B,L}}]^T \in \mathbb{R}_+^{L \times 1}$. In order to ensure the system performance and user fairness, the NOMA-weak user should be allocated more power than that of the NOMA-strong user [38], thereby the transmit power for Roy and Bob should satisfy that

$$\beta_l = \frac{P_{B,l}}{P_R} \in [0, 1), \quad \forall l \in \mathcal{L}, \quad (6)$$

where β_l is the power allocation ratio of the l -th LED. Therefore, the electrical signal in the k -th symbol transmitted from the l -th LED can be expressed as

$$x_l[k] = \begin{cases} \sqrt{P_R}u_R[k], & \mathcal{H}_0, \\ \sqrt{P_R}u_R[k] + \sqrt{P_{B,l}}u_B[k], & \mathcal{H}_1, \end{cases} \quad (7)$$

where the null hypothesis $\mathcal{H}_0$ states that LEDs did not transmit the covert information, while the alternative hypothesis $\mathcal{H}_1$ implies that LEDs did transmit the covert information.

For VLCC systems, the electrical signal $x_l[k]$ is converted by electrical-optical conversion into an optical signal $X_l[k] = \varsigma(x_l[k] + D)$, where ς is the electrical-to-optical conversion coefficient, and D is a direct current (DC) bias to ensure the derived optical signal is non-negative. Next, the optical signal arrives at the receiver through both the LoS and OIRS-reflected links, and undergoes optical-to-electrical conversion. After the DC bias is removed, the derived electrical signal at receiver $i \in \{R, B, W\}$ can be written as

$$y_i[k] = \begin{cases} \varsigma\mu\mathbf{g}_i^T \mathbf{w}_R u_R[k] + z_i[k], & \mathcal{H}_0, \\ \varsigma\mu\mathbf{g}_i^T \mathbf{w}_R u_R[k] + \varsigma\mu\mathbf{g}_i^T \mathbf{w}_B u_B[k] + z_i[k], & \mathcal{H}_1, \end{cases} \quad (8)$$

where μ is the optical-to-electrical conversion coefficient. $z_i[k] \sim \mathcal{N}(0, \sigma_i^2)$ is the zero-mean Gaussian noise at receiver i with variance of σ_i^2 , including ambient light fluctuations and hardware-induced variations.

III. PERFORMANCE ANALYSIS OF THE VLCC SYSTEM

In this section, the detection performance of Willie is discussed in Section III-A, including his decision threshold and the actual DEP. Then, the COPs of the users Roy and Bob are analyzed in Section III-B.

A. Detection Performance of Willie

For Willie, we first consider the limited-knowledge case where it lacks the knowledge of OIRS state, thereby revealing the optimal system performance. Therefore, Willie needs to decide the decision threshold λ based on the CSI of LoS link $\mathbf{h}_W$. Moreover, when Willie's LoS channel is unavailable, the

system controller may prevent Willie from monitoring by not assigning OIRS to it. Consequently, the LoS channel constitutes the fundamental prerequisite for Willie's monitoring.

Since sophisticated detectors like generalized likelihood ratio test (GLRT) and machine learning (ML)-based classifiers offer advantages under system uncertainty [39], [40], they are not suitable for the proposed VLCC system, as it is assumed that Willie possesses perfect knowledge of signal distributions. Therefore, Willie can employ an energy detector to compare the average received power against a predetermined threshold λ , which is given by

$$T_W = \frac{1}{K} \sum_{k=1}^K |y_W[k]|^2 \stackrel{\mathcal{D}_0}{\underset{\mathcal{D}_1}{\leq}} \lambda, \quad (9)$$

where $\mathcal{D}_0$ and $\mathcal{D}_1$ are the decisions in favor of $\mathcal{H}_0$ and $\mathcal{H}_1$, respectively.

Moreover, the average signal power that Willie thinks he is receiving through the LoS link $\mathbf{h}_W$ can be expressed as

$$\tilde{T}_W = \begin{cases} \varsigma^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_R + \sigma_W^2, & \mathcal{H}_0, \\ \varsigma^2 \mu^2 \left(\|\mathbf{h}_W\|_1^2 + (\mathbf{h}_W^T \sqrt{\boldsymbol{\beta}})^2 \right) P_R + \sigma_W^2, & \mathcal{H}_1, \end{cases} \quad (10)$$

where $\sqrt{\boldsymbol{\beta}} = [\sqrt{\beta_1}, \dots, \sqrt{\beta_L}]^T \in \mathbb{R}_+^{L \times 1}$. Under the assumption of equal priori probabilities of hypotheses $\mathcal{H}_0$ and $\mathcal{H}_1$, the detection performance of Willie can be measured by its DEP ξ , which is defined as

$$\xi = \mathcal{P}_{\text{FA}} + \mathcal{P}_{\text{MD}}, \quad (11)$$

where $\mathcal{P}_{\text{FA}} = \Pr(\mathcal{D}_1 | \mathcal{H}_0)$ is the false alarm probability, which means that Willie judged that LEDs transmitted the covert information, while in fact LEDs did not transmit it. $\mathcal{P}_{\text{MD}} = \Pr(\mathcal{D}_0 | \mathcal{H}_1)$ is the miss detection probability, which means that Willie concluded that LEDs did not transmit the covert information, while actually LEDs did transmit it.

First, the perceived false alarm probability that Willie believes in can be written as

$$\begin{aligned} \tilde{\mathcal{P}}_{\text{FA}} &= \Pr \left(\varsigma^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_R + \sigma_W^2 \geq \lambda \mid \mathcal{H}_0 \right) \\ &= \begin{cases} 1, & \lambda < \sigma_W^2, \\ 1 - \frac{\lambda - \sigma_W^2}{\varsigma^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_{\text{max}}}, & \sigma_W^2 \leq \lambda \leq \varpi_{\text{FA}}, \\ 0, & \lambda > \varpi_{\text{FA}}, \end{cases} \quad (12) \end{aligned}$$

where $\varpi_{\text{FA}} = \varsigma^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_{\text{max}} + \sigma_W^2$. Then, the perceived miss detection probability that Willie believes in is given by

$$\begin{aligned} \tilde{\mathcal{P}}_{\text{MD}} &= \Pr \left(\varsigma^2 \mu^2 \Upsilon_{\mathbf{h}_W} P_R + \sigma_W^2 \leq \lambda \mid \mathcal{H}_1 \right) \\ &= \begin{cases} 0, & \lambda < \sigma_W^2 \\ \frac{\lambda - \sigma_W^2}{\varsigma^2 \mu^2 \Upsilon_{\mathbf{h}_W} P_{\text{max}}}, & \sigma_W^2 \leq \lambda \leq \varpi_{\text{MD}} \\ 1, & \lambda > \varpi_{\text{MD}} \end{cases} \quad (13) \end{aligned}$$

where $\Upsilon_{\mathbf{h}_W} = \|\mathbf{h}_W\|_1^2 + (\mathbf{h}_W^T \sqrt{\boldsymbol{\beta}})^2$ and $\varpi_{\text{MD}} = \varsigma^2 \mu^2 \Upsilon_{\mathbf{h}_W} P_{\text{max}} + \sigma_W^2$. Moreover, noting that $\Upsilon_{\mathbf{h}_W} \geq \|\mathbf{h}_W\|_1^2$

makes $\varpi_{MD} \geq \varpi_{FA}$, the perceived DEP that Willie believes in can be presented by

$$\tilde{\xi}(\lambda) = \begin{cases} 1, & \lambda < \sigma_W^2, \\ 1 - \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_{\max}}, & \sigma_W^2 \leq \lambda \leq \varpi_{FA}, \\ \frac{\lambda - \sigma_W^2}{\zeta^2 \mu^2 \Upsilon_{\mathbf{h}_W} P_{\max}}, & \varpi_{FA} < \lambda \leq \varpi_{MD}, \\ 1, & \lambda > \varpi_{MD}. \end{cases} \quad (14)$$

For Willie, he needs to choose the optimal decision threshold λ^* that minimizes his perceived DEP $\tilde{\xi}(\lambda)$ according to the CSI of LoS links. Observation of $\tilde{\xi}(\lambda)$ shows that when $\sigma_W^2 \leq \lambda \leq \varpi_{FA}$, the value of $\tilde{\xi}(\lambda)$ decreases as λ increases, while when $\varpi_{FA} < \lambda \leq \varpi_{MD}$, the value of $\tilde{\xi}(\lambda)$ increases as λ increases. In summary, Willie would set the optimal decision threshold λ^* as

$$\lambda^* = \varpi_{FA} = \zeta^2 \mu^2 \|\mathbf{h}_W\|_1^2 P_{\max} + \sigma_W^2. \quad (15)$$

However, the detection of Willie can be confused through the OIRS-reflected link $\mathbf{q}_W$, thereby increasing its true DEP value. In reality, the average signal power distribution received by Willie through both the LoS link $\mathbf{h}_W$ and OIRS-reflected link $\mathbf{q}_W$ can be given by

$$T_W = \begin{cases} \zeta^2 \mu^2 \|\mathbf{g}_W\|_1^2 P_R + \sigma_W^2, & \mathcal{H}_0, \\ \zeta^2 \mu^2 (\|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2) P_R + \sigma_W^2, & \mathcal{H}_1. \end{cases} \quad (16)$$

Therefore, the actual false alarm probability at Willie can be given by

$$\begin{aligned} \mathcal{P}_{FA} &= \Pr(\zeta^2 \mu^2 \|\mathbf{g}_W\|_1^2 P_R + \sigma_W^2 \geq \lambda^* \mid \mathcal{H}_0) \\ &= 1 - \frac{\|\mathbf{h}_W\|_1^2}{\|\mathbf{g}_W\|_1^2}. \end{aligned} \quad (17)$$

Similarly, the actual miss detection probability at Willie can be obtained as

$$\mathcal{P}_{MD} = \Pr(\zeta^2 \mu^2 \Upsilon_{\mathbf{g}_W} P_R + \sigma_W^2 \leq \lambda^* \mid \mathcal{H}_1) = \frac{\|\mathbf{h}_W\|_1^2}{\Upsilon_{\mathbf{g}_W}} \quad (18)$$

where $\Upsilon_{\mathbf{g}_W} = \|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2$. Consequently, the actual DEP at Willie can be expressed as

$$\xi^* = 1 - \frac{\|\mathbf{h}_W\|_1^2}{\|\mathbf{g}_W\|_1^2} + \frac{\|\mathbf{h}_W\|_1^2}{\Upsilon_{\mathbf{g}_W}}. \quad (19)$$

B. Connection Outage Probability

In order to achieve the covert communication, the transmit power is set to a random varying value. However, when the transmit power is too small, it may lead to interruption of communication for legitimate users Roy and Bob, thus reducing the communication efficiency. Therefore, in order to balance the covert communication requirements and the connection outage probability, it is necessary to reasonably optimize the power allocation and OIRS configuration.

In VLCC systems, the relationship between the SE and the generalized signal-to-interference-and-noise ratio (SINR) can be expressed as [41]

$$R_i = \frac{1}{2} \log_2(1 + \gamma_i), \quad i \in \{B \rightarrow R, B, R\}. \quad (20)$$

As Bob and Roy are legitimate users of the VLCC system, they differ from Willie in their ability to obtain the complete CSI through communications with the system controller. According to the SIC process in NOMA, Bob needs to first demodulate the information of Roy with higher power, and then remove it before demodulating his own message. As a result, the generalized SINR for Bob to decode Roy's signal $u_R[k]$ can be given by

$$\gamma_{B \rightarrow R} = \frac{\zeta^2 \mu^2 (\mathbf{g}_B^T \mathbf{w}_R)^2}{\zeta^2 \mu^2 (\mathbf{g}_B^T \mathbf{w}_B)^2 + \tilde{\sigma}_B^2}, \quad (21)$$

where $\tilde{\sigma}_B^2 = e\pi\sigma_B^2/2$ denotes the generalized noise of Bob [41]. After perfectly removing the signal of Roy [7], Bob will only receive the effect of Gaussian noise when he detects his own signal, and the generalized SINR can be written as

$$\gamma_B = \frac{\zeta^2 \mu^2 (\mathbf{g}_B^T \mathbf{w}_B)^2}{\tilde{\sigma}_B^2}. \quad (22)$$

Furthermore, when Roy demodulates his own signal, the signal of Bob is treated as interference along with the Gaussian noise, so that the generalized SINR can be expressed as

$$\gamma_R = \frac{\zeta^2 \mu^2 (\mathbf{g}_R^T \mathbf{w}_R)^2}{\zeta^2 \mu^2 (\mathbf{g}_R^T \mathbf{w}_B)^2 + \tilde{\sigma}_R^2}, \quad (23)$$

where $\tilde{\sigma}_R^2 = e\pi\sigma_R^2/2$ denotes the generalized noise of Roy. When the LoS channel is unavailable, i.e., only the OIRS-reflected path exists, the aforementioned analysis still holds by setting the LoS channel gain $\mathbf{h}_i$ to zero.

Based on the previous discussion, the COP of the covert user Bob for the NOMA-based VLCC system can be obtained by

$$\begin{aligned} \Xi_B &= 1 - \Pr(R_{B \rightarrow R} \geq R_R^{\text{th}}, R_B \geq R_B^{\text{th}}) \\ &= 1 - \Pr(\gamma_{B \rightarrow R} \geq \alpha_R, \gamma_B \geq \alpha_B), \end{aligned} \quad (24)$$

where R_R^{th} and R_B^{th} are the pre-determined SEs for Roy and Bob, respectively, and $\alpha_R = 2^{2R_R^{\text{th}}} - 1$, $\alpha_B = 2^{2R_B^{\text{th}}} - 1$. Similarly, the COP of the user Roy is given by

$$\Xi_R = \Pr(R_R < R_R^{\text{th}}) = \Pr(\gamma_R < \alpha_R). \quad (25)$$

Lemma 1: The expressions for the COPs of Roy and Bob can be written as

$$\Xi_R = \begin{cases} \frac{\alpha_R \tilde{\sigma}_R^2 / \zeta^2 \mu^2 P_{\max}}{\|\mathbf{g}_R\|_1^2 - \alpha_R (\mathbf{g}_R^T \sqrt{\beta})^2}, & (\mathbf{g}_R^T \sqrt{\beta})^2 < \frac{\|\mathbf{g}_R\|_1^2}{\alpha_R}, \\ 1, & (\mathbf{g}_R^T \sqrt{\beta})^2 \geq \frac{\|\mathbf{g}_R\|_1^2}{\alpha_R}, \end{cases} \quad (26)$$

$$\Xi_B = \begin{cases} 1, & (\mathbf{g}_B^T \sqrt{\beta})^2 \leq \varpi_1, \\ \frac{\alpha_B \tilde{\sigma}_B^2}{\zeta^2 \mu^2 (\mathbf{g}_B^T \sqrt{\beta})^2 P_{\max}}, & \varpi_1 < (\mathbf{g}_B^T \sqrt{\beta})^2 \leq \varpi_2, \\ \frac{\alpha_R \tilde{\sigma}_B^2 / \zeta^2 \mu^2 P_{\max}}{\|\mathbf{g}_B\|_1^2 - \alpha_R (\mathbf{g}_B^T \sqrt{\beta})^2}, & \varpi_2 < (\mathbf{g}_B^T \sqrt{\beta})^2 < \varpi_3, \\ 1, & (\mathbf{g}_B^T \sqrt{\beta})^2 \geq \varpi_3, \end{cases} \quad (27)$$

where ϖ_1 , ϖ_2 and ϖ_3 are given by

$$\varpi_1 = \frac{\alpha_B \tilde{\sigma}_B^2}{\zeta^2 \mu^2 \|\mathbf{g}_B\|_1^2 P_{\max}}, \quad (28)$$

$$\varpi_2 = \frac{\alpha_B \|\mathbf{g}_B\|_1^2}{\alpha_R + \alpha_R \alpha_B}, \quad (29)$$

$$\varpi_3 = \frac{\|\mathbf{g}_B\|_1^2}{\alpha_R}. \quad (30)$$

Proof: The derivation is given in Appendix A. ■

IV. PROBLEM FORMULATION AND OPTIMIZATION

In this section, the optimization problem for the NOMA-based VLCC system with OIRS in MISO scenario is formulated in Section IV-A. Then, the power allocation and the OIRS configuration sub-problems are addressed separately in Sections IV-B and IV-C, respectively. The complete process of the BCD algorithm is presented in Section IV-D. Moreover, the convergence and computational complexity of the proposed algorithm are also discussed in Section IV-E.

A. Problem Formulation

Given the COP of Bob Ξ_B , the effective covert SE of the VLCC system can be expressed as $R_B^{\text{th}}(1 - \Xi_B)$. Therefore, the problem of maximizing the effective covert SE by optimizing the power allocation ratio vector β and the OIRS configuration matrix $\mathbf{S}$ can be given by

$$(\mathbf{P0}) : \max_{\mathbf{S}, \beta} R_B^{\text{th}}(1 - \Xi_B) \quad (31)$$

$$\text{s.t.} \quad \xi^* \geq 1 - \kappa, \quad (32)$$

$$\Xi_R \leq \Xi_R^{\text{th}}, \quad (33)$$

$$s_{l,n,i} \in \{0, 1\}, \quad \forall l \in \mathcal{L}, n \in \mathcal{N}, i \in \{B, R, W\}, \quad (34)$$

$$\sum_{l=1}^L s_{l,n,B} + s_{l,n,R} + s_{l,n,W} = 1, \quad \forall n \in \mathcal{N}, \quad (35)$$

$$0 \leq \beta_l < 1, \quad \forall l \in \mathcal{L}. \quad (36)$$

Constraint in (32) is the DEP constraint to ensure the transmission of covert information between LEDs and the covert user Bob. Then, constraint in (33) is a COP constraint, where Ξ_R^{th} is the value of the maximum COP allowed by the public user Roy to ensure that the information transmission of Roy can proceed normally. Moreover, constraints in (34) and (35) are derived based on the definition of the OIRS configuration matrix $\mathbf{S}$, since each OIRS unit can only be assigned to one LED-PD pair. Besides, constraint in (36) limits the transmit power for Bob to be no larger than that for Roy.

Observing the expression for DEP ξ^* in (19), the constraint in (32) can be equivalently transformed into

$$(\mathbf{g}_W^T \sqrt{\beta})^2 \leq \frac{\kappa \|\mathbf{g}_W\|_1^4}{\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2}, \quad (37)$$

where the equivalence of the above transformation is that $\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2$ always guarantees a value larger than zero. When $\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2 \leq 0$, the DEP constraint is always satisfied, thus the constraint in (37) no longer needs to be considered. Moreover, if condition $\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2 \leq 0$ is held, then it means that fewer OIRS units can be allocated to Willie to make $\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2$ larger than zero. At this point, the DEP constraint in (37) can still be easily satisfied, and the saved OIRS units can be assigned to Bob to improve his effective covert SE. Therefore, the number of OIRS units allocated to Willie should always not be too large, i.e., $\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2$ is always larger than zero.

Similarly, the constraint in (33) can also be equivalently transformed according to (26), which can be rewritten as

$$(\mathbf{g}_R^T \sqrt{\beta})^2 \leq \frac{\|\mathbf{g}_R\|_1^2}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\zeta^2 \mu^2 P_{\max} \Xi_R^{\text{th}}}. \quad (38)$$

Furthermore, the objective function in (31) is equivalent to minimize the COP of Bob Ξ_B . Observing the expression of the COP for Bob in (27), it can be seen that when $\varpi_{B1} < (\mathbf{g}_B^T \sqrt{\beta})^2 \leq \varpi_{B2}$, Ξ_B decreases as $(\mathbf{g}_B^T \sqrt{\beta})^2$ increases, and when $\varpi_{B2} < (\mathbf{g}_B^T \sqrt{\beta})^2 < \varpi_{B3}$, Ξ_B increases as $(\mathbf{g}_B^T \sqrt{\beta})^2$ increases. Therefore, in order to minimize the COP of Bob, it should be satisfied that

$$(\mathbf{g}_B^T \sqrt{\beta})^2 \leq \frac{\alpha_B \|\mathbf{g}_B\|_1^2}{\alpha_R + \alpha_R \alpha_B}. \quad (39)$$

In addition, it can be observed that maximizing the effective covert SE of Bob is equivalent to maximizing $(\mathbf{g}_B^T \sqrt{\beta})^2$. Since $\mathbf{g}_B^T \sqrt{\beta}$ is always greater than zero, the original problem (P0) can be transformed into the equivalent form as

$$(\mathbf{P1}) : \max_{\mathbf{S}, \beta} \mathbf{g}_B^T \sqrt{\beta} \quad (40)$$

s.t. (34), (35), (36), (37), (38), (39).

However, the variables $\mathbf{S}$ and β interact with each other in the problem (P1), it is difficult to solve them directly. Therefore, the BCD algorithm is utilized to decompose the original problem into two sub-problems, and solve each sub-problem individually to maximize the objective function in (40).

B. Power Allocation

In this subsection, the power ratio vector β is optimized with the fixed OIRS configuration matrix $\mathbf{S}$. Therefore, the power allocation optimization sub-problem can be given by

$$(\mathbf{P2}) : \max_{\beta} \mathbf{g}_B^T \sqrt{\beta} \quad (41)$$

s.t. (36), (37), (38), (39).

Notice that both the objective function and the constraints are about $\sqrt{\beta}$, which can be considered as a new variable

to be optimized. Thus, the power allocation optimization sub-problem can be rewritten as

$$\begin{aligned} (\mathbf{P2-1}) : & \max_{\sqrt{\beta}} \mathbf{g}_B^T \sqrt{\beta} \\ \text{s.t.} \quad & 0 \leq \sqrt{\beta_l} < 1, \quad \forall l \in \mathcal{L}, \end{aligned} \quad (42)$$

$$(37), (38), (39). \quad (43)$$

After transforming the optimization variable, it can be found that the objective function is linear with respect to $\sqrt{\beta}$, the constraint in (43) is also a linear constraint, and the constraints in (37), (38) and (39) are all convex constraints. As a result, the power allocation sub-problem (P2-1) can be solved by the convex optimization method, and the interior point method (IPM) is applied in this paper [42]. After solving by IPM, the final power allocation ratio vector β can be obtained directly from $\sqrt{\beta}$, since $\sqrt{\beta}$ and β have a one-to-one correspondence with respect to the interval $[0, 1)$.

C. OIRS Configuration

In this subsection, the OIRS configuration matrix $\mathbf{S}$ is iteratively optimized with the fixed power allocation ratio vector β . Thereby the OIRS configuration sub-problem can be formulated as

$$\begin{aligned} (\mathbf{P3}) : & \max_{\mathbf{S}} \mathbf{g}_B^T \sqrt{\beta} \\ \text{s.t.} \quad & (34), (35), (37), (38), (39). \end{aligned} \quad (44)$$

It should be noted that although the objective function in (44) is linear with respect to the OIRS configuration matrix $\mathbf{S}$, all the constraints in (34), (37), (38), and (39) are non-convex. In particular, the integer constraint in (34) leads to the sub-problem (P3) being a mixed-integer non-linear program (MINLP), which is difficult to solve. Therefore, the integer constraint in (34) is first relaxed into a continuous form to facilitate the subsequent solution [26], which can be given by

$$0 \leq s_{l,n,i} \leq 1, \quad \forall l \in \mathcal{L}, n \in \mathcal{N}, i \in \{B, R, W\}. \quad (45)$$

Moreover, notice that both the left and right sides of the constraint (37) are real numbers larger than zero, thus the expression can be simplified by taking logarithms of both left and right sides, leading to the equivalent form as

$$2\ln \mathbf{g}_W^T \sqrt{\beta} + \ln \left(\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2 \right) \leq \ln \kappa + 4\ln \|\mathbf{g}_W\|_1, \quad (46)$$

where $\mathbf{g}_W$ is a linear function about $\mathbf{S}$. Therefore, the left and right sides of (46) become concave functions.

In order to make the constraint in (46) a convex constraint that is easy to solve in each round of iterations, the MM algorithm is adopted to relax the constraint and solve the sub-problem (P3) iteratively [43]. Firstly, the following two upper bound functions and an additional constraint are given as

$$2\ln \mathbf{g}_W^T \sqrt{\beta} \leq 2\ln \mathbf{g}_W^{*T} \sqrt{\beta} + \frac{2(\mathbf{g}_W^T - \mathbf{g}_W^{*T}) \sqrt{\beta}}{\mathbf{g}_W^{*T} \sqrt{\beta}}, \quad (47a)$$

$$\begin{aligned} \ln \left(\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2 \right) & \leq \ln \left(\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W^*\|_1^2 \right) \\ & - \frac{2\kappa \|\mathbf{g}_W^*\|_1 (\|\mathbf{g}_W\|_1 - \|\mathbf{g}_W^*\|_1)}{\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W^*\|_1^2}, \end{aligned} \quad (47b)$$

$$\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W\|_1^2 > 0, \quad (48)$$

where $\mathbf{g}_W^*$ denotes a fixed value calculated from the result $\mathbf{S}^{(r)}$ in the r -th round of iterations, and the constraint (48) is utilized to ensure the validity of the logarithmic function in (46). Consequently, based on the above upper bounds, the constraint (46) can be relaxed as

$$\begin{aligned} 2\ln \mathbf{g}_W^{*T} \sqrt{\beta} + \frac{2(\mathbf{g}_W^T - \mathbf{g}_W^{*T}) \sqrt{\beta}}{\mathbf{g}_W^{*T} \sqrt{\beta}} + \ln \left(\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W^*\|_1^2 \right) \\ - \frac{2\kappa \|\mathbf{g}_W^*\|_1 (\|\mathbf{g}_W\|_1 - \|\mathbf{g}_W^*\|_1)}{\|\mathbf{h}_W\|_1^2 - \kappa \|\mathbf{g}_W^*\|_1^2} \leq \ln \kappa + 4\ln \|\mathbf{g}_W\|_1, \end{aligned} \quad (49)$$

where the left side is a linear function while the right side is a concave function with respect to $\mathbf{S}$, thereby the relaxed constraint in (49) becomes a convex constraint.

Similarly, for the non-convex constraints in (38) and (39), the Taylor expansion is first utilized to obtain the following two lower bound functions as

$$\|\mathbf{g}_R\|_1^2 \geq \underline{\|\mathbf{g}_R\|_1^2} = \|\mathbf{g}_R^*\|_1^2 + 2\|\mathbf{g}_R^*\|_1 (\|\mathbf{g}_R\|_1 - \|\mathbf{g}_R^*\|_1), \quad (50)$$

$$\|\mathbf{g}_B\|_1^2 \geq \underline{\|\mathbf{g}_B\|_1^2} = \|\mathbf{g}_B^*\|_1^2 + 2\|\mathbf{g}_B^*\|_1 (\|\mathbf{g}_B\|_1 - \|\mathbf{g}_B^*\|_1), \quad (51)$$

where both $\mathbf{g}_R^*$ and $\mathbf{g}_B^*$ are fixed values that are calculated from the result $\mathbf{S}^{(r)}$ in the r -th round of iterations. Then, by utilizing the linear functions $\underline{\|\mathbf{g}_B\|_1^2}$ and $\underline{\|\mathbf{g}_R\|_1^2}$ instead of $\|\mathbf{g}_B\|_1^2$ and $\|\mathbf{g}_R\|_1^2$ in constraints (38) and (39), respectively, the following relaxed constraints can be obtained as

$$\left(\mathbf{g}_R^T \sqrt{\beta} \right)^2 \leq \frac{\|\mathbf{g}_R\|_1^2}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\zeta^2 \mu^2 P_{\max} \Xi_R^{\text{th}}}, \quad (52)$$

$$\left(\mathbf{g}_B^T \sqrt{\beta} \right)^2 \leq \frac{\alpha_B \|\mathbf{g}_B\|_1^2}{\alpha_R + \alpha_R \alpha_B}. \quad (53)$$

Similarly, the relaxed constraints in (52) and (53) becomes convex, and the original sub-problem (P3) can be relaxed into

$$\begin{aligned} (\mathbf{P3-1}) : & \max_{\mathbf{S}} \mathbf{g}_B^T \sqrt{\beta} \\ \text{s.t.} \quad & (35), (45), (48a), (47), (51), (52), \end{aligned} \quad (54)$$

where the objective function and all constraints are convex, so that can also be solved by IPM [42]. Moreover, the integer constraint (34) needs to be recovered from the optimized result $\tilde{\mathbf{S}}^{(r)}$, where the minimum distance criterion is adopted [27]. Specifically, for the n -th OIRS unit, the maximum value in $\tilde{s}_{l,n,i}^{(r)}$, $\forall l \in \mathcal{L}, i \in \{R, B, W\}$ is assigned to one and the other values are set to zero. Although this recovery method incurs a slight performance loss, it is justified by a substantial reduction in computational complexity. Relevant research confirms that such a trade-off is highly favorable, as the performance degradation remains within acceptable limits while the complexity savings are considerable [26].

D. BCD Algorithm for the Problem (P0)

To be specific, the original problem (P0) is decomposed into two sub-problems to solve β and $\mathbf{S}$ iteratively according

Algorithm 1 BCD Algorithm for the Effective Covert SE Maximization Problem (P0)

```

1: Init: iteration rounds  $r \leftarrow 0$ , the convergence threshold  $\epsilon$ ,
   the initial  $\Xi_B^{(0)}$  calculated with  $\mathbf{S}^{(0)}$  and  $\beta^{(0)}$ .
2: Output:  $\mathbf{S}^*$  and  $\beta^*$ .
3: repeat
4:    $r \leftarrow r + 1$ ;
5:   Given  $\mathbf{S}^{(r-1)}$ , solve the sub-problem (P2-1) by IPM,
   and obtain the power allocation ratio vector  $\beta^{(r)}$ ;
6:   Given  $\beta^{(r)}$ , solve the relaxed sub-problem (P3-1) to
   obtain the OIRS configuration matrix  $\tilde{\mathbf{S}}^{(r)}$ ;
7:    $n \leftarrow 1$ ;
8:   repeat
9:      $l^*, i^* \leftarrow \arg \max_{l,i} \tilde{s}_{l,n,i}^{(r)} s_{l^*,n,i^*}^{(r)} \leftarrow 1$ ;
10:     $s_{l',n,i'}^{(r)} \leftarrow 0, l' \neq l^* \text{ or } i' \neq i^*$ ;
11:     $n \leftarrow n + 1$ ;
12:  until  $n > N$ ;
13:  Calculate  $\Xi_B^{(r)}$  based on  $\mathbf{S}^{(r)}$  and  $\beta^{(r)}$ ;
14: until  $|R_B^{\text{th}}(\Xi_B^{(r)} - \Xi_B^{(r-1)})| \leq \epsilon$ ;
15:  $\mathbf{S}^* \leftarrow \mathbf{S}^{(r)}, \beta^* \leftarrow \beta^{(r)}$ .

```

to the BCD algorithm. Firstly, given the OIRS configuration matrix $\mathbf{S}$, the square root of the power ratio vector $\sqrt{\beta}$ can be obtained by solving the sub-problem (P2-1), and then β can be calculated by $\sqrt{\beta}$. After that, the OIRS configuration sub-problem (P3-1) can be solved by the relaxed MM algorithm with fixed β , then the final OIRS configuration matrix $\mathbf{S}$ can be recovered by the minimum distance criterion. Within each round of the iteration, the effective covert SE of the VLCC system is calculated based on the latest variables $\mathbf{S}$ and β , and compared with the results of the previous round. When the difference between the effective covert SE of two adjacent iteration rounds is less than the error threshold ϵ , the iteration is ended and the final optimization results $\mathbf{S}^*$ and β^* are output. To sum up, the complete algorithmic process is described in **Algorithm 1**.

E. Analysis of Convergence and Computational Complexity

To analyze the convergence of the proposed BCD algorithm for the effective covert SE maximization in the MISO scenario, the following three aspects should be considered.

First, the sub-problems (P2) and (P3) are relaxed to (P2-1) and (P3-1), respectively, utilizing the MM algorithm, i.e., the original non-convex functions is replaced with strict convex constraints by utilizing upper or lower bound functions, so that the results $\mathbf{S}$ and β obtained from the final solution can still satisfy the original constraints. For instance, the inequalities in (47), (50), and (51) not only ensure that the relaxed constraints in (49), (52), and (53) are convex constraints, but also makes the solved result satisfy the original constraints (37), (38), and (39) as well.

Second, since the convex optimization algorithm IPM is utilized to solve the sub-problems in each iteration, it can be guaranteed that the effective covert SE increases after each

iteration, i.e.,

$$R_B^{\text{th}} \left(1 - \Xi_B^{(r)}\right) \geq R_B^{\text{th}} \left(1 - \Xi_B^{(r-1)}\right), \quad \forall r \geq 1, \quad (55)$$

which ensures that the objective function will always trend upwards (or at least not downwards) without oscillatory situations leading to non-convergence.

Third, the variables to be optimized, i.e., $\mathbf{S}$ and β , are subject to both upper and lower bound constraints in (34) and (36). Therefore, these two optimization variables will not increase without limit, thus ensuring that the iterative process can eventually converge.

For the MISO scenario, the complexity of solving the two sub-problems separately will be discussed before giving the total complexity.

1) *Power Allocation for (P2)*: Since the dimension of the power allocation ratio vector β is L , the complexity of IPM is $\mathcal{O}(L^{3.5} \log_2(1/\eta_1))$ [44], where η_1 represents the solution accuracy of IPM.

2) *OIRS Configuration for (P3)*: The solution process for the sub-problem (P3) consists of solving the relaxed sub-problem (P3-1) and the recovery of the integer constraint in (34). Given the dimension of $\mathbf{S}$, the complexity of solving (P3-1) by IPM with the solution accuracy η_2 is $\mathcal{O}((3LN)^{3.5} \log_2(1/\eta_2))$ [44]. For the recovery of the integer constraint, the computational complexity of search operations is $\mathcal{O}(3LN)$, since at most $3L$ searches are required for each OIRS unit. To sum up, the computational complexity is $\mathcal{O}((3LN)^{3.5} \log_2(1/\eta_2) + 3LN)$.

As Algorithm 1 alternately solves the sub-problems (P2) and (P3), the number of iterations is approximately $\mathcal{O}(\log_2(1/\epsilon))$ [42]. Therefore, the total computational complexity of the proposed BCD algorithm to solve the problem (P0) is $\mathcal{O}((L^{3.5} \log_2(1/\eta_1) + (3LN)^{3.5} \log_2(1/\eta_2) + 3LN) \log_2(1/\epsilon))$.

V. ENHANCED ADVERSARY CASE AND IMPERFECT CSI

In this section, further factors detrimental to the VLCC system are examined. Specifically, we first discuss the evolution of covert communication performance when confronting a more powerful adversary in Section V-A, meaning that Willie possesses knowledge of OIRS state at any given moment. Secondly, the imperfect CSI acquisition of Willie is also addressed in Section V-B.

A. Willie With Knowledge of OIRS State

With access to the real-time OIRS configuration, Willie gains superior monitoring capabilities, and the distribution of received power it perceives reflects the precise statistical profile of the signal. Therefore, Willie can obtain the average received power as

$$T'_W = \begin{cases} \varsigma^2 \mu^2 \|\mathbf{g}_W\|_1^2 P_R + \sigma_W^2, & \mathcal{H}_0, \\ \varsigma^2 \mu^2 \left(\|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2 \right) P_R + \sigma_W^2, & \mathcal{H}_1. \end{cases} \quad (56)$$

Then, Willie can derive the genuinely optimal decision threshold through a reasoning process analogous to that in (14), which can be expressed as

$$\lambda' = \varsigma^2 \mu^2 \|\mathbf{g}_W\|_1^2 P_{\max} + \sigma_W^2. \quad (57)$$

Based on this decision threshold in (57) and the received power distribution in (56), the false alarm probability and miss detection probability for Willie can be given by

$$\mathcal{P}'_{\text{FA}} = \Pr\left(\zeta^2 \mu^2 \|\mathbf{g}_W\|_1^2 P_R + \sigma_W^2 \geq \lambda' \mid \mathcal{H}_0\right) = 0, \quad (58)$$

$$\begin{aligned} \mathcal{P}'_{\text{MD}} &= \Pr\left(\zeta^2 \mu^2 \Upsilon_{\mathbf{g}_W} P_R + \sigma_W^2 \leq \lambda' \mid \mathcal{H}_1\right) \\ &= \frac{\|\mathbf{g}_W\|_1^2}{\|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2}. \end{aligned} \quad (59)$$

Consequently, by directly adding the values of $\mathcal{P}'_{\text{FA}}$ and $\mathcal{P}'_{\text{MD}}$, the DEP can be obtained as

$$\xi' = \frac{\|\mathbf{g}_W\|_1^2}{\|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2}. \quad (60)$$

It should be noted that the enhanced capabilities of Willie solely affect the value of his DEP, with no impact on the two legitimate users, Bob and Roy. Therefore, the remaining discussions in Section III remain applicable for this case. The new optimization problem can thus be formulated as

$$(\mathbf{P0}'): \max_{\mathbf{S}, \beta} R_B^{\text{th}} (1 - \Xi_B) \quad (61)$$

$$\begin{aligned} \text{s.t.} \quad & \xi' \geq 1 - \kappa, \\ & (33), (34), (35), (36). \end{aligned} \quad (62)$$

where the only difference from the problem (P0) lies in constraint (62), and it can be further transformed as

$$(\mathbf{g}_W^T \sqrt{\beta})^2 \leq \frac{\kappa}{1 - \kappa} \|\mathbf{g}_W\|_1^2, \quad (63)$$

which is more concise than the transformed constraint in (37). Therefore, the method employed to solve the problem (P0) can likewise be applied to solve the problem (P0'). Specifically, when addressing the power allocation sub-problem, it constitutes a convex optimization problem that can be solved directly. When tackling the OIRS configuration sub-problem, constraint (63) can be equivalently transformed to

$$2\ln \mathbf{g}_W^{*T} \sqrt{\beta} + \frac{2(\mathbf{g}_W^T - \mathbf{g}_W^{*T}) \sqrt{\beta}}{\mathbf{g}_W^{*T} \sqrt{\beta}} \leq \ln \frac{\kappa}{1 - \kappa} + 2\ln \|\mathbf{g}_W\|_1. \quad (64)$$

Subsequently, the solution to the OIRS configuration sub-problem remains consistent, and it can be resolved utilizing the methodology employed in the sub-problem (P3).

B. Imperfect CSI of Willie

In VLCC systems, as the warden Willie is an unfriendly user, the system controller may be unable to obtain the accurate CSI of Willie. In such circumstances, the imperfect CSI of Willie may also adversely affect the covert communication system [13], [45], necessitating a discussion.

To begin with, let the true LoS channel gain of Willie be denoted as $\mathbf{h}_W$, and the estimated value as $\tilde{\mathbf{h}}_W$. When Willie's CSI is imperfect, it can be assumed that a bias $\Delta \mathbf{h}_W = \mathbf{h}_W - \tilde{\mathbf{h}}_W$ exists between the true and estimated values, and this bias is bounded by $|\Delta \mathbf{h}_W| \leq \tau \mathbf{h}_W$, where $0 \leq \tau \leq 1$ is a coefficient utilized to quantify the CSI estimation error [45]. Similarly, the true and estimated values

of the OIRS-reflected channel gain for Willie can be denoted as $\mathbf{f}_{l,W}$ and $\tilde{\mathbf{f}}_{l,W}$, respectively, and the relationship between them satisfies that $|\Delta \mathbf{f}_{l,W}| = |\mathbf{f}_{l,W} - \tilde{\mathbf{f}}_{l,W}| \leq \tau \mathbf{f}_{l,W}$. Moreover, it is worth noting that this CSI uncertainty can also be regarded as the diffuse reflection model.

Then, the imperfect CSI of Willie leads to an inaccurate DEP value as perceived by the system controller [45]. Taking the instance where Willie lacks the knowledge of OIRS state, the true value of its DEP should be written as

$$\xi^* = \frac{\|\mathbf{h}_W\|_1^2}{\|\mathbf{g}_W\|_1^2 + (\mathbf{g}_W^T \sqrt{\beta})^2} + 1 - \frac{\|\mathbf{h}_W\|_1^2}{\|\mathbf{g}_W\|_1^2}. \quad (65)$$

Therefore, to ensure that the DEP constraint remains satisfied under actual Willie channel conditions, we must incorporate Willie's estimated channel gain $\tilde{\mathbf{g}}_W$ and bias coefficient τ into (65), thereby guaranteeing that the true channel gain value also satisfies the DEP constraint as

$$\begin{aligned} & \frac{\|\tilde{\mathbf{h}}_W + \Delta \mathbf{h}_W\|_1^2}{\|\tilde{\mathbf{g}}_W + \Delta \mathbf{g}_W\|_1^2 + ((\tilde{\mathbf{g}}_W + \Delta \mathbf{g}_W)^T \sqrt{\beta})^2} \\ & + 1 - \frac{\|\tilde{\mathbf{h}}_W + \Delta \mathbf{h}_W\|_1^2}{\|\tilde{\mathbf{g}}_W + \Delta \mathbf{g}_W\|_1^2} \geq 1 - \kappa, \end{aligned} \quad (66)$$

where $\Delta \mathbf{g}_W$ can be calculated by $\Delta \mathbf{h}_W$ and $\Delta \mathbf{f}_{l,W}$. Moreover, regarding Willie's knowledge of the OIRS configuration, both $\tilde{\mathbf{g}}_W$ and τ must be substituted into (60), ensuring that the actual CSI value remains within the deviation range to satisfy that $\xi' \geq 1 - \kappa$.

VI. SPECIAL CASE FOR SISO SCENARIO

In this section, the issue of covert communications in SISO scenarios is also discussed, and the closed-form solution for the power allocation ratio is derived, which means that only the OIRS configuration needs to be optimized in Section VI-A. Moreover, the convergence and computational complexity are also analyzed in Section VI-B.

A. Power Allocation and OIRS Configuration

In the SISO scenario, there is only one LED at the transmitter side, i.e., $L = 1$. As a result, the power allocation ratio changes from the vector β for the MISO scenario to a scalar β , the OIRS configuration matrix becomes $\mathbf{S} \in \mathbb{B}^{N \times 3}$, and the total channel gain between the LED and Bob, Roy, and Willie for the SISO scenario can also be represented by the scalars g_B , g_R , and g_W , respectively.

Lemma 2: When Willie lacks the knowledge of OIRS state, the optimal power allocation ratio can be obtained as

$$\begin{aligned} \beta^* &= \min \left\{ \frac{\kappa g_W^2}{h_W^2 - \kappa g_W^2}, \frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\zeta^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}}, \right. \\ & \left. \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B}, 1 \right\}, \end{aligned} \quad (67)$$

which means that the optimal power ratio should be the smallest one of the four values in (67). When Willie possesses knowledge of OIRS state, the first item in (67) becomes $\kappa/(1 - \kappa)$, with the remainder unchanged.

Proof: The derivation is given in Appendix B. ■

Since the power allocation ratio β^* can be written in the form about the OIRS configuration matrix $\mathbf{S}$, the problem of maximizing the effective covert SE for the SISO scenario can be transformed into an optimization problem that only focused on the OIRS configuration, which can be formulated as

$$\begin{aligned} (\mathbf{P-A}) : & \max_{\mathbf{S}} g_B^2 \beta^* \\ \text{s.t.} & (34), (35). \end{aligned} \quad (68)$$

Similarly, it is first necessary to relax the integer constraint (34) to a continuous form, i.e.

$$0 \leq s_{1,n,i} \leq 1, \quad \forall n \in \mathcal{N}, i \in \{B, R, W\}. \quad (69)$$

Bringing β^* in (67) into the objective function in (68) reveals that the problem to be solved is a max-min problem, and then (P-A) can be transformed into an equivalent form by introducing a new optimization variable Γ as

$$(\mathbf{P-B}) : \max_{\mathbf{S}, \Gamma} \Gamma \quad (70)$$

$$\text{s.t.} \quad 2\ln g_B + 2\ln \sqrt{\kappa} g_W - \ln(h_W^2 - \kappa g_W^2) \geq \Gamma, \quad (71)$$

$$2\ln g_B + \ln \left(\frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}} \right) \geq \Gamma, \quad (72)$$

$$2\ln g_B + \ln \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B} \geq \Gamma, \quad (73)$$

$$2\ln g_B \geq \Gamma, \quad (74)$$

$$(35), (68),$$

where the left sides of constraints (71), (72), (73), and (74) are taken logarithmically to simplify the expression, since the value of $g_B^2 \beta^*$ is always greater than zero.

However, constraints (71) and (72) in the problem (P-B) are still non-convex, which need to be solved by the MM algorithm iteratively. In each round of iterations, these constraints should be transformed into convex forms to find the upper and lower bounds. Specifically, the following inequalities can be obtained based on Taylor expansion as

$$\begin{aligned} \ln(h_W^2 - \kappa g_W^2) &\leq \ln(h_W^2 - \kappa (g_W^*)^2) \\ &\quad - \frac{2\kappa g_W^*}{h_W^2 - \kappa (g_W^*)^2} (g_W - g_W^*), \end{aligned} \quad (75)$$

$$g_R^2 \geq \underline{g}_R^2 = (g_R^*)^2 + 2g_R^* (g_R - g_R^*), \quad (76)$$

where g_W^* and g_R^* are fixed values that are calculated from the result $\mathbf{S}^{(r)}$ in the r -th round of iterations. Consequently, the problem (P-B) can be relaxed as

$$(\mathbf{P-C}) : \max_{\mathbf{S}, \Gamma} \Gamma \quad (77)$$

$$\begin{aligned} \text{s.t.} \quad & 2\ln g_B + 2\ln \sqrt{\kappa} g_W \geq \Gamma + \ln(h_W^2 - \kappa (g_W^*)^2) \\ & \quad - \frac{2\kappa g_W^* (g_W - g_W^*)}{h_W^2 - \kappa (g_W^*)^2}, \end{aligned} \quad (78)$$

$$2\ln g_B + \ln \left(\frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 \underline{g}_R^2 P_{\max} \Xi_R^{\text{th}}} \right) \geq \Gamma, \quad (79)$$

$$(35), (68), (72), (73),$$

where the objective function in (77) is linear, and all constraints are convex with respect to $\mathbf{S}$ and Γ . Therefore, the

problem (P-C) can be solved iteratively by the convex optimization method, e.g. IPM [42]. In each round of iteration, the integer constraint in (35) can be recovered by the minimum distance criterion just like solving the sub-problem (P3-1). The above iterative process will continue until the effective covert SE difference between two adjacent iterations is less than a threshold value ϵ to obtain the final OIRS configuration matrix $\mathbf{S}^*$, and the value of the power allocation ratio β^* can be calculated according to (67).

B. Analysis of Convergence and Computational Complexity

For the convergence analysis in the SISO scenario, it can be easily shown that the three conditions for the MISO scenario are also satisfied in the iterative process, and therefore the optimization process for the effective covert SE in the SISO scenario is also convergent.

For the computational complexity in the SISO scenario, we only need to analyze the complexity of solving the OIRS configuration, because the power ratio β can be given directly as an analytic without optimization. Since the dimension of the OIRS configuration matrix $\mathbf{S}$ is $3N$ in the SISO scenario, the complexity of solving the sub-problem (P-C) by IPM in each iteration is $\mathcal{O}((3N)^{3.5} \log_2(1/\eta))$ [44]. Recovering the integer constraint requires $\mathcal{O}(3N)$ operations in each round of iteration. The convergence threshold of the whole algorithm is ϵ , thereby the total complexity of the algorithm in the SISO scenario is $\mathcal{O}(((3N)^{3.5} \log_2(1/\eta) + 3N) \log_2(1/\epsilon))$. It can be seen that the computational complexity in the SISO scenario is significantly lower than that in the MISO scenario.

VII. NUMERICAL RESULTS

In this section, simulation results are presented for the OIRS-assisted VLCC system with NOMA. The convergence of the proposed algorithm is shown in Section VII-A, followed by the performance of the algorithms for solving the sub-problems compared against other baselines. Moreover, the impact of key parameters on system performance is provided in Section VII-B.

The dimensions of the room are set to 4 m $\times$ 4 m $\times$ 3 m. The number of LEDs L is 4, and LEDs are arranged in a square with 1 meter apart [35]. In addition, an OIRS consisting of 16 or 64 units is arranged in a 4 $\times$ 4 or 8 $\times$ 8 configuration on a wall, with the point (0 m, 2 m, 2 m) as the center. The area of each OIRS unit is set to the appropriate size 10 $\times$ 10 cm², while each OIRS unit is spaced 5 cm apart to ensure that each unit can be freely adjusted [27]. Besides, other important parameters are listed in Table I [27], [29], and the parameters will not be changed in the sequel unless there are special instructions.

A. Simulations of the Proposed Algorithm

The convergence performance of the proposed algorithm during the iterations is shown in Fig. 2, where each iteration represents solving the sub-problems (P2) and (P3). It can be observed that the algorithm requires approximately 4 iterations to converge. In addition, by comparing the converged effective covert SE, it is evident that the proposed algorithm has a

TABLE II
SIMULATION PARAMETERS

Parameter	Value
The number of LEDs, L	4
The number of OIRS units, N	16 or 64
The Lambertian index, m	1
The semi-angle of the FOV, Φ_0	80°
The area of PD, A	1 cm^2
The internal refractive index, b	1.5
Electrical-to-optical conversion factor, ς	0.5
Optical-to-electrical conversion factor, μ	0.5
The reflection factor, δ	0.95
Pre-determined SE, $R_R^{\text{th}}, R_B^{\text{th}}$	0.5, 1 bps/Hz
DEP constraint, κ	0.25
COP constraint of Roy, Ξ_R^{th}	0.5
Transmit power limit, $P_{\max}$	7.5 W

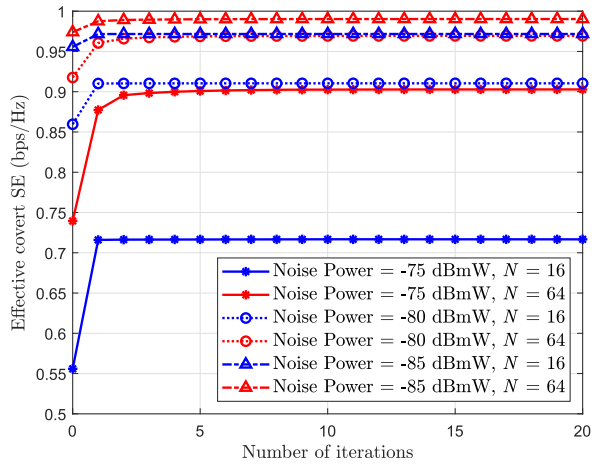


Fig. 2. Convergence of the proposed algorithm for solving (P0) under different noise powers and number of OIRS units.

very significant improvement compared to the initial state when the noise power is high. Taking the noise power equals to -75 dBmW , $N = 16$ as an example, the proposed algorithm increases the SE from 0.55 bps/Hz initially to almost 0.72 bps/Hz , significantly enhance the effective covert SE of Bob. Furthermore, the improvement in the effective covert SE by increasing the number of OIRS units is also more pronounced when the noise power is high, indicating that OIRS is effective for VLCC systems in low SNR environments.

Moreover, the proposed algorithms for solving the sub-problems are compared with the following benchmark schemes with 64 OIRS units.

- **Unified power allocation [12]:** The power allocation ratio of each LED is restricted to be the same value, that is, $\beta_i = \beta_l, \forall i, l \in \mathcal{L}$.
- **Fixed power allocation [15]:** The power allocation ratio of each LED is fixed to the medium value 0.5 , i.e., $\beta_l = 0.5, \forall l \in \mathcal{L}$, and only the OIRS configuration matrix $\mathbf{S}$ is optimized.

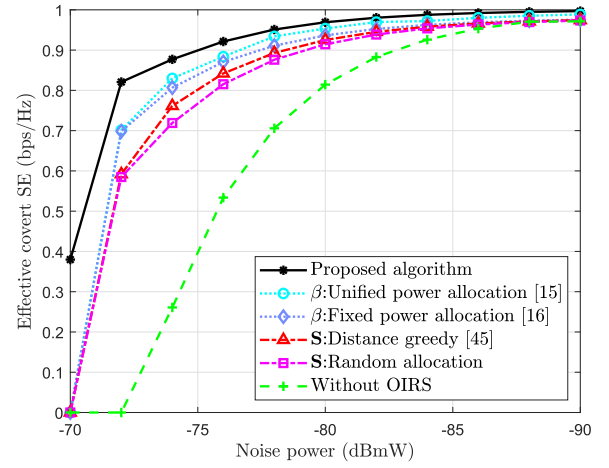


Fig. 3. The effective covert SE versus the noise power compared to different benchmark schemes.

- **Distance greedy [46]:** In this case, each OIRS unit is aligned greedily, which means each unit is assigned to the closest LED and user $i \in \{B, R, W\}$, and only the power allocation ratio vector is optimized.
- **Random allocation:** The OIRS configuration is determined by the random assignment of each unit, only the power allocation sub-problem is performed.
- **Without OIRS:** There is no OIRS in the room, so that the proposed algorithm just needs to solve the power allocation sub-problem.

As shown in Fig. 3, the proposed BCD algorithm achieves the best performance and has a nearly 0.4 bps/Hz improvement over the second-best performance baseline when the noise power is -70 dBmW . Moreover, the diversity of the power allocation ratio between multiple LEDs also provides a performance boost to the VLCC system. In addition, comparing the optimization of the OIRS configuration matrix $\mathbf{S}$ and the power allocation ratio vector β , it can be seen that OIRS configuration improves the system performance more significantly, especially when the noise power is high. The reason is that when the noise power σ^2 is high, the COP for Bob and Roy decreases more markedly with the introduction of OIRS. This outcome aligns with the findings presented in (26) and (27). In particular, if there is no OIRS in the room, the effective covert SE of Bob remains low even when the power allocation ratio is optimized, which proves that the introduction of OIRS has a very important role in the enhancement of effective covert SE.

The simulation results concerning the capability of Willie to possess knowledge of OIRS state and its CSI status under different noise powers are presented in Fig. 4, where the number of OIRS units is 64 and the bias coefficient τ for imperfect CSI is 0.2. It can be observed that when Willie is aware of the OIRS configuration, the effective covert SE experiences a slight decline at low SNR. This occurs because the effect of OIRS on Willie is diminished at this point. However, compared to the scenario without OIRS, the values are higher when Willie is aware of the OIRS state. This indicates that even when facing a powerful monitor, the incorporation of OIRS substantially

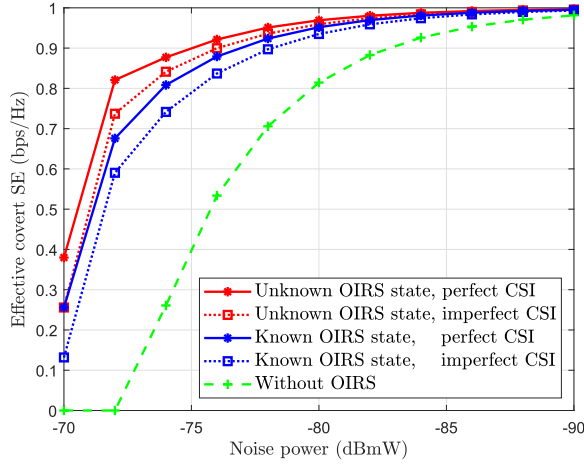


Fig. 4. The effective covert SE versus the capability of Willie and its CSI status under different noise powers.

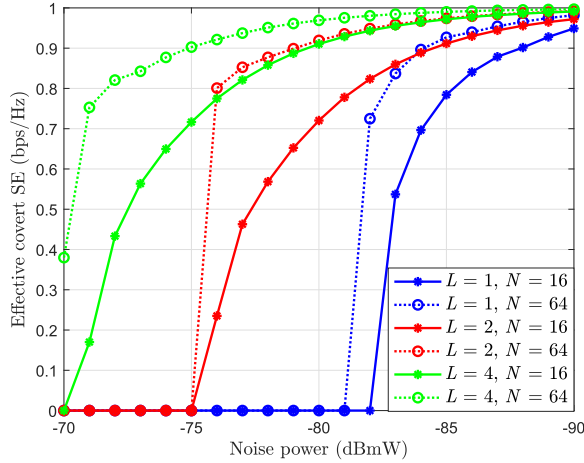


Fig. 5. The effective covert SE versus the noise power with different number of LEDs and OIRS units.

enhances the effective covert SE. Moreover, the CSI deviation of Willie also causes some deterioration on the effective covert SE, as it requires ensuring that all scenarios within the CSI deviation regions satisfy the DEP constraint, thereby wasting a portion of the resources of the VLCC system.

B. Simulations on Key Parameters

Fig. 5 illustrates the effective covert SE versus the noise power with different number of LEDs and OIRS units. For different numbers of LEDs, all LEDs are in the center of the ceiling area and 1 meter away from each other. It can be found that when there are fewer LEDs and higher noise power, the effective covert SE of Bob is all zero. This is because the total transmit power is too small at this time, and to satisfy the DEP restriction and the COP constraint of Roy, the power allocation ratio is limited to a very small value, which leads to the interruption of Bob's communication. However, for different numbers of LEDs, the effective covert SEs of the VLCC system all undergo a sudden surge from zero around a certain noise power, and this surge is more pronounced with a larger number of OIRS units, which suggests that the

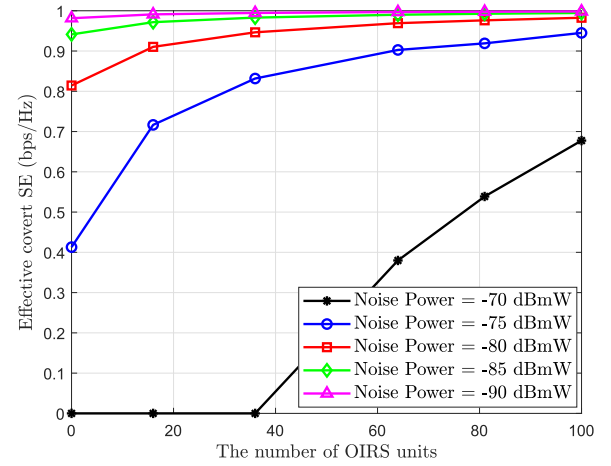


Fig. 6. The effective covert SE versus the number of OIRS units with different noise powers.

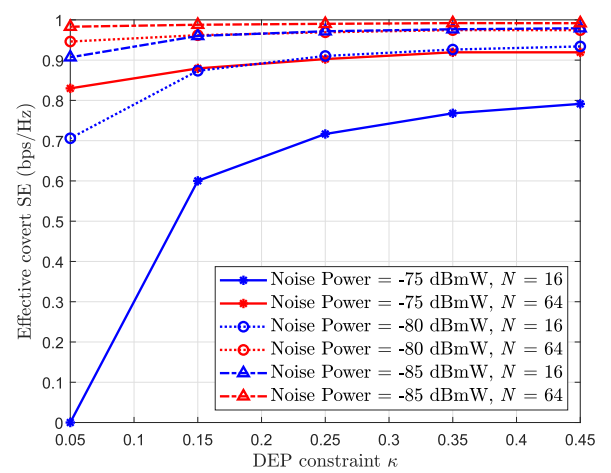


Fig. 7. The effective covert SE versus the DEP constraint factor k with different noise powers and number of OIRS units.

performance of VLCC systems can be improved significantly by OIRS around the threshold value of noise power.

Fig. 6 shows the improvement of the effective covert SE by varying the number of OIRS units under different noise powers. It should be noted that adding more OIRS units results in remarkable enhancement in the effective covert SE, and the increase is more pronounced at higher noise power, which means that increasing the number of OIRS units can assist the VLCC system to combat noise. In particular, when the noise power is -75 dBmW, the effective covert SE is 0.95 bps/Hz and 0.40 bps/Hz for 100 OIRS units and no OIRS, respectively, which is a performance improvement of more than two times. Notably, since R_B^{th} is set to 1 bps/Hz, the maximum value of effective covert SE is 1 bps/Hz, which indicates that 100 OIRS units can achieve a superior performance even if the noise power is very high.

The effective covert SE versus the DEP constraint factor κ with different noise powers and number of OIRS units is shown in Fig. 7. When the DEP constraint is very strict, i.e., $\kappa = 0.05$, the effective covert SE is generally low at this point, due to the fact that the power allocation ratio for Bob has

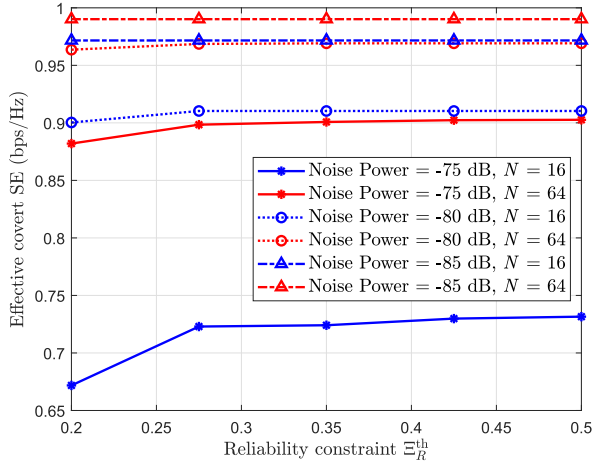


Fig. 8. The effective covert SE versus the reliability constraint factor Ξ_R^{th} with different noise powers and number of OIRS units.

to be sufficiently small to satisfy the covertness requirement, leading to easy interruption of the Bob's communication. However, increasing the number of OIRS units can significantly mitigate this phenomenon. Taking the noise power is -75 dBmW, $\kappa = 0.05$ as an example, adding the OIRS units results in a gain of 0.7 bps/Hz, which is because OIRS not only helps to satisfy the covert communication demand, but also improves the channel quality of Bob thus reducing his COP.

In Fig. 8, the effect of the COP constraint factor Ξ_R^{th} on the effective covert SE is shown with different noise powers and number of OIRS units. It is worth noting that the effect of Ξ_R^{th} on effective covert SE of Bob is similar to that of the DEP constraint factor κ . This is because the smaller the values of factors κ and Ξ_R^{th} , the tighter the constraints become, thus limiting the value of β and leading to a higher COP for Bob. Moreover, when the noise power is small, changing the COP constraint factor Ξ_R^{th} hardly affects the effective covert SE, suggesting that the COP constraint can be easily satisfied and is no longer a determinant of the effective covert SE performance under high SNR conditions.

VIII. CONCLUSION

In this paper, an optimization architecture of the OIRS-assisted VLCC system with NOMA was established with the effective covert SE as the optimization objective. To address the original problem in the MISO scenario, it was decomposed into two sub-problems, i.e., the power allocation and OIRS configuration, and then the BCD algorithm was employed to solve the sub-problems iteratively. For the power allocation sub-problem, it can be solved directly utilizing convex optimization by means of an equivalent transformation. For the OIRS configuration sub-problem, the integer constraint was first relaxed, and then it was solved by the MM algorithm. In the SISO scenario, the closed-form solutions for the power allocation ratio were derived, reducing the complexity of the problem. Numerical results demonstrated that the proposed algorithms converge fast and achieves superior performance compared to other baselines, especially when the noise power

is high. Moreover, the simulation results showed that the increase in the number of OIRS units and LEDs can bring significant growth to the effective covert SE. In summary, OIRS can enhance the effective covert SE of NOMA-based VLCC systems, and the technology of OIRS shows great potential to improve the covertness performance for future optical communications.

APPENDIX A PROOF OF LEMMA 1

For the COP expression of Roy, an equivalent transformation can be obtained as

$$\begin{aligned}\Xi_R &= \Pr \left(\frac{\varsigma^2 \mu^2 \|\mathbf{g}_R\|_1^2 P_R}{\varsigma^2 \mu^2 (\mathbf{g}_R^T \sqrt{\beta})^2 P_R + \tilde{\sigma}_R^2} < \alpha_R \right) \\ &= \Pr \left(\frac{1}{P_R} > \frac{\|\mathbf{g}_R\|_1^2 - \alpha_R (\mathbf{g}_R^T \sqrt{\beta})^2}{\alpha_R \tilde{\sigma}_R^2 / \varsigma^2 \mu^2} \right). \quad (80)\end{aligned}$$

When $\|\mathbf{g}_R\|_1^2 - \alpha_R (\mathbf{g}_R^T \sqrt{\beta})^2 \leq 0$, the value of the right side in (80) is less than 0, so the value of Ξ_R is 1. Besides, when $\|\mathbf{g}_R\|_1^2 - \alpha_R (\mathbf{g}_R^T \sqrt{\beta})^2 > 0$, the expression in (26) can be obtained from the definition of uniform distribution.

For the COP expression of Bob, when $(\mathbf{g}_B^T \sqrt{\beta})^2 \geq \varpi_3$, $\gamma_{B \rightarrow R} \leq \alpha_R$ always holds, thus $\Xi_B = 1$. Similarly, when $(\mathbf{g}_B^T \sqrt{\beta})^2 \leq \varpi_1$, $\gamma_B \leq \alpha_B$ is satisfied, thereby $\Xi_B = 1$.

Moreover, it is worth noting that when $\mathbf{g}_B^T \sqrt{\beta}$ increases, the SINR that Bob decodes his own signal increases while the SINR that Bob decodes Roy's signal decreases, and so the value of the cutoff ϖ_2 can be obtained by solving for $\gamma_{B \rightarrow R} = \alpha_R$ and $\gamma_B = \alpha_B$ jointly. Therefore, when $\varpi_1 < (\mathbf{g}_B^T \sqrt{\beta})^2 \leq \varpi_2$, the COP of Bob can be obtained by

$$\Xi_B = \Pr(\gamma_B < \alpha_B) = \Pr \left(P_R < \frac{\alpha_B \tilde{\sigma}_B^2}{\varsigma^2 \mu^2 (\mathbf{g}_B^T \sqrt{\beta})^2} \right). \quad (81)$$

Furthermore, when $\varpi_2 < (\mathbf{g}_B^T \sqrt{\beta})^2 < \varpi_3$, the COP of Bob can be given by

$$\begin{aligned}\Xi_B &= \Pr(\gamma_{B \rightarrow R} < \alpha_R) \\ &= \Pr \left(P_R < \frac{\alpha_R \tilde{\sigma}_B^2 / \varsigma^2 \mu^2}{\|\mathbf{g}_B\|_1^2 - \alpha_R (\mathbf{g}_B^T \sqrt{\beta})^2} \right). \quad (82)\end{aligned}$$

In summary, combining all of the above intervals can yield a complete expression for the COP of Bob in (27).

APPENDIX B PROOF OF LEMMA 2

First, when Willie is unaware of the OIRS configuration, the actual DEP for the SISO scenario can be expressed as $\xi^* = 1 - \beta h_W^2 / (1 + \beta) g_W^2$, and thereby the DEP constraint in (32) can be transformed into the equivalent form as

$$\beta \leq \frac{\kappa g_W^2}{h_W^2 - \kappa g_W^2}. \quad (83)$$

When Willie knows the OIRS configuration, the DEP can be given by $\xi' = 1 - \beta / (1 + \beta)$, and thus the DEP constraint

can be transformed to $\beta \leq \kappa/(1 - \kappa)$. Moreover, constraint in (33) for the SISO scenario can be rewritten as

$$\beta \leq \frac{1}{\alpha_R} - \frac{\tilde{\sigma}_R^2}{\varsigma^2 \mu^2 g_R^2 P_{\max} \Xi_R^{\text{th}}}, \quad (84)$$

which is obtained from the expression for the COP of Roy in (26) under the SISO scenario. Furthermore, according to the COP of Bob in (27), the power allocation ratio should satisfy that

$$\beta \leq \frac{\alpha_B}{\alpha_R + \alpha_R \alpha_B}. \quad (85)$$

Notice that β should also satisfy the NOMA power allocation constraint as

$$0 \leq \beta < 1. \quad (86)$$

Therefore, in order to maximize the effective covert SE, i.e., to maximize β while satisfying the constraints, the value of β should be the smallest of the four previous inequalities in (83), (84), (85), and (86).

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